

Ready-to-Eat Dried Meat

Required HACCP / ISO 22000 Additions for Manual, Records, Validation, and System Forms

Review document only. No system or manual changes are implemented yet. This document assumes the dried meat is manufactured internally and consumed without cooking after drying.

1. Auditor Conclusion

Because the product is Ready-to-Eat (RTE), drying and seasoning cannot be treated only as a quality process. The FSMS must prove that hazards are controlled before product release, especially where no later cooking step exists.

- Current dried meat record is a strong starting point, but it needs formal QA release, validation, and post-drying contamination controls.
- The HACCP manual should include a dedicated RTE Dried Meat product description, process flow, hazard analysis, and HACCP/OPRP plan.
- Product release should be blocked unless required limits, calibration checks, lab references, label checks, and QA approval are complete.

2. Critical Decision Points Before Implementation

Decision	What TELT Must Confirm	Why It Matters
Product classification	Confirm final classification as Ready-to-Eat Dried Meat.	Defines stronger hygiene, release, testing, and label controls.
Lethality step	Confirm whether drying has validated pathogen reduction, or only reduces water activity.	If there is no validated kill step, raw material and post-process controls become critical.
Storage condition	Confirm shelf-stable, chilled, or frozen storage based on validated aw, pH, moisture, packaging, and shelf-life.	Incorrect storage statement creates legal and safety risk.
Critical limits	Approve target limits for aw, pH, moisture, salt, nitrite/nitrate, time, temperature, and humidity.	Limits must be validated and legally compliant, not only convenient values in the form.
Testing plan	Approve frequency for microbiology, environmental swabs, retained samples, and shelf-life verification.	Provides evidence that RTE product remains safe through shelf life.

3. Required HACCP Manual Amendments

Add or update the following sections in the FSMS / HACCP manual.

Manual Section	Required Addition
Scope	State that the FSMS covers in-house manufacturing of Ready-to-Eat Dried Meat, not only chilled/frozen meat handling.
Product Description	Add product name, raw meat type, ingredients, spices, additives, allergens, packaging, intended use, consumer group, storage condition, shelf life, and distribution method.
Process Flow	Create a dedicated flow: receiving, storage, thawing if applicable, trimming/slicing, weighing/formulation, curing, drying, cooling, testing, packaging, labeling, storage, dispatch.
Onsite Flow Verification	Add RTE Dried Meat to the verified flow list with date, team members, deviations found, and approval.
Hazard Analysis	Analyze each step for biological, chemical, physical, and allergen hazards. Include Salmonella, Listeria, E. coli, Staph aureus toxin risk, molds, nitrite/nitrate, cleaning chemicals, bone, metal, and packaging fragments.
Control Measure Categorization	Categorize drying endpoint, curing formulation, chilled curing hold, packaging hygiene, label verification, and storage as PRP, OPRP, or CCP based on decision tree.
HACCP / OPRP Plan	Define limits, monitoring method, frequency, responsible person, correction, corrective action, verification, and record for each significant control measure.
Validation	Add evidence for aw/pH/moisture limits, drying time-temperature profile, shelf-life study, microbiology testing, and packaging/storage conditions.
Potentially Unsafe Product	State that any batch missing release tests, exceeding limits, or exposed to post-drying contamination is held until evaluated by QA.
Legal Register Link	Reference applicable UAE/GSO/municipality rules for dried meat, additives, labeling, shelf-life, microbiological criteria, and storage declarations.

4. Product Specification Additions

Specification Field	Required Content
Product identity	Ready-to-Eat Dried Meat, product variant, recipe/version number.
Raw material	Species, supplier, lot/SIF, halal status, receiving temperature, acceptance criteria.
Ingredients/additives	Spices, salt, nitrite/nitrate if used, allergens, ingredient lots, approved limits.
Food safety limits	aw target and maximum, pH target, moisture target, salt content, nitrite/nitrate legal limit if applicable.
Packaging	Vacuum/MAP/regular pack, food-contact approval, seal requirements, pack size.
Label	RTE statement if used, storage condition, production/expiry, allergens, preservatives, country/origin requirements, lot/batch code.
Shelf life	Approved shelf life with validation reference and storage condition.
Release criteria	No release until QA approval, required tests complete, and deviations closed.

5. Required Records and Reports

Record / Report	Purpose	Minimum Fields
Raw Material Receiving Record	Prove approved and acceptable raw meat.	Supplier, lot/SIF, halal, temperature, production/expiry, COA, acceptance/rejection.
Defrosting Record	Required if frozen raw meat is used.	Batch, start/end time, room/product temperature, responsible person, disposition.
Pre-Operation Sanitation Record	Prevent RTE contamination from equipment and environment.	Area/equipment, chemical, concentration, visual check, ATP/swab if used, QA verification.
Cutting / Slicing Check	Control physical hazards and hygiene before curing.	Blade/knife condition, bone/foreign matter check, area temperature, operator.
Recipe / Formula Weighing Record	Control salt, curing agents, spices, and allergens.	Recipe version, ingredient lot, target weight, actual weight, second-person verification.
Curing Time/Temperature Record	Control growth before drying.	Start/end, curing temp, product temp, duration, deviations, corrective action.
Dried Meat Process Record	Main batch manufacturing record.	Drying log, temperature, humidity, time, position, weight loss, aw, pH, moisture, salt, sensory, packaging.
Water Activity Meter Verification	Prove aw result reliability.	Meter ID, standard used, expected value, actual value, pass/fail, performed by.
pH Meter Calibration Record	Prove pH result reliability.	Meter ID, buffers used, readings, slope if applicable, pass/fail, performed by.
Final Product Release Record	Prevent unsafe product from sale.	Batch, test results, label check, retained sample, QA decision, release/hold/reject reason.
Microbiology Test Report Log	Verify RTE safety.	Sample ID, batch, lab, Salmonella, Listeria, E. coli/Staph if applicable, result, action.
Environmental Monitoring / Swab Log	Detect contamination risk in drying/packing area.	Zone, surface, organism/indicator, result, trend, corrective action.
Packaging and Label Verification	Prevent legal and allergen errors.	Label version, lot/date, allergens, preservatives, storage, expiry, barcode, checked by.
Retained Sample Log	Support complaints, shelf-life verification, and investigations.	Batch, sample count, storage condition, retention until, disposal date.
Traceability and Mass Balance	Prove raw-to-finished link.	Raw lots, input weight, final weight, yield, pack count, destination.
Nonconformance / CAPA	Control deviations.	Deviation, affected batch, hold status, root cause, correction, preventive action, closure.
Shelf-Life Verification Log	Confirm product remains safe and acceptable.	Batch, storage condition, test date, sensory, aw/pH/micro results, decision.
Training Record	Prove operator competence.	RTE hygiene, curing formula, aw/pH testing, allergen/label controls, cleaning.

6. Required System/Form Additions

Add these fields to the existing Dried Meat Process module or create linked sub-records.

Area	Fields to Add
Batch Control	Recipe Version, Product Specification ID, Production Line/Room, Operator, QA Reviewer.
Formula Control	Salt Target/Actual, Nitrite/Nitrate Target/Actual, Spice Lot Numbers, Allergen Review, Formula Checked By.
Critical Limits	Approved aw Limit, Approved pH Limit, Moisture Limit, Salt Limit, Drying Temperature Limit, Curing Temperature Limit.
Equipment Verification	aw Meter ID, aw Meter Verification Result, pH Meter ID, pH Calibration Result, Thermometer ID.
Microbiology	Lab Sample Ref, Lab Name, Test Panel, Result Status, Date Sent, Date Received.
Environmental Monitoring	Swab Ref, Surface/Zone, Result, Corrective Action Required.
Packaging/Label	Packaging Lot, Seal Check, Label Version, Storage Statement Checked, Allergen/Preservative Checked, Label Checked By.
Release	QA Release Decision: Released / Hold / Rejected, Hold Reason, Corrective Action, QA Release By, Release Date.
Retention	Retained Sample ID, Quantity, Storage Condition, Retention Until.

7. Suggested Release Logic

Condition	System Action
aw result missing	Block release. Batch remains Hold.
aw above approved limit	Block release. Require QA evaluation and CAPA.
pH or moisture missing where required by specification	Block release until completed.
Calibration not verified	Block release or require QA override with documented justification.
Micro result pending for high-risk/validation batches	Hold until result is acceptable.
Label check incomplete	Block release.
Any post-drying contamination event	Hold affected batch and require investigation.
All limits acceptable and QA signs	Allow Released status and lock release record.

8. Validation and Verification Requirements

Activity	Minimum Expectation	Frequency
Initial process validation	Prove selected drying time/temperature/humidity and aw/pH/moisture produce safe RTE dried meat.	Before full approval and after major changes.
Shelf-life validation	Support expiry date under actual storage and packaging.	Before label approval; repeat after formulation/process/package change.
Microbiology verification	Finished product testing for relevant RTE pathogens/indicators.	Risk-based; increase during validation and after deviations.
Environmental monitoring	Swabs in drying, cooling, slicing, and packaging areas.	Routine schedule, trend monthly.
Calibration/verification	aw meter, pH meter, thermometer, weighing scales.	Before use or as defined by calibration plan.
Internal audit check	Review completed batch records, release decisions, CAPA, and traceability.	At least annually, preferably quarterly for new RTE process.
Mock recall	Include RTE dried meat batch in traceability drill.	At least annually or per company recall program.

9. Implementation Priority

Recommended order before adding everything into the system:

- Priority 1: Add QA release decision, hold/reject logic, calibration checks, and label verification to the existing dried meat form.
- Priority 2: Add manual amendments: product description, flow, hazard analysis, HACCP/OPRP plan, and validation statement.
- Priority 3: Add linked records for aw meter, pH meter, environmental swabs, microbiology, retained samples, and shelf-life verification.
- Priority 4: Add dashboard indicators for batches on Hold, released batches, failed limits, pending lab results, and overdue verification.

10. References for Review

- Codex CXC 1-1969: General Principles of Food Hygiene, including HACCP principles.
- Codex CXC 75-2015: Code of Hygienic Practice for Low-Moisture Foods.
- FDA Food Code 2022: reference model for food safety definitions and controls.
- Applicable UAE, GSO, Dubai Municipality, Abu Dhabi, and MoCCA requirements must be confirmed in the Legal Register before final limits are approved.